hw_13

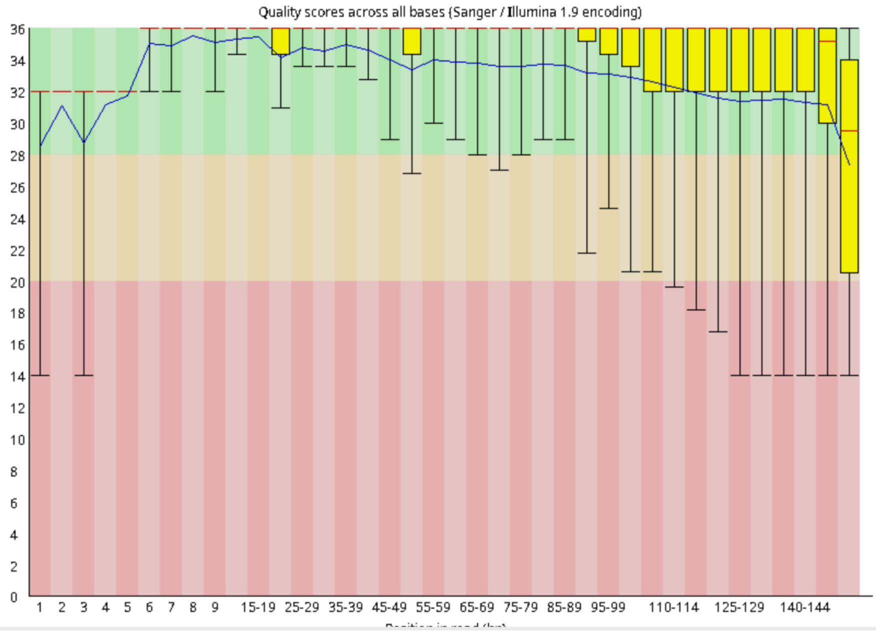
R1

Summary

Basic Statistics

Measure	Value
Filename	Erik_mat_CD4_MiLab_S52_R1_001.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1983876
Total Bases	291.3 Mbp
Sequences flagged as poor quality	0
Sequence length	35-151
%GC	41

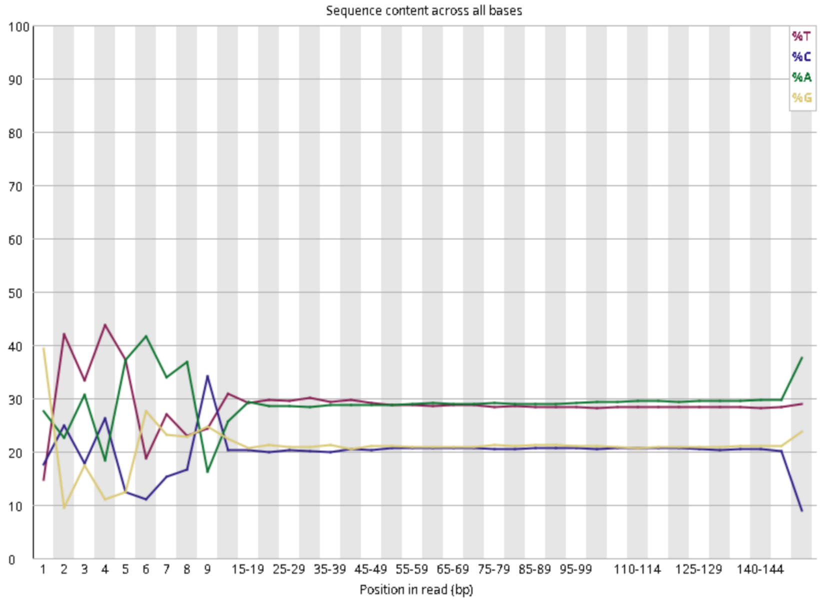
Per base sequence quality



Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

Per base sequence content



R2

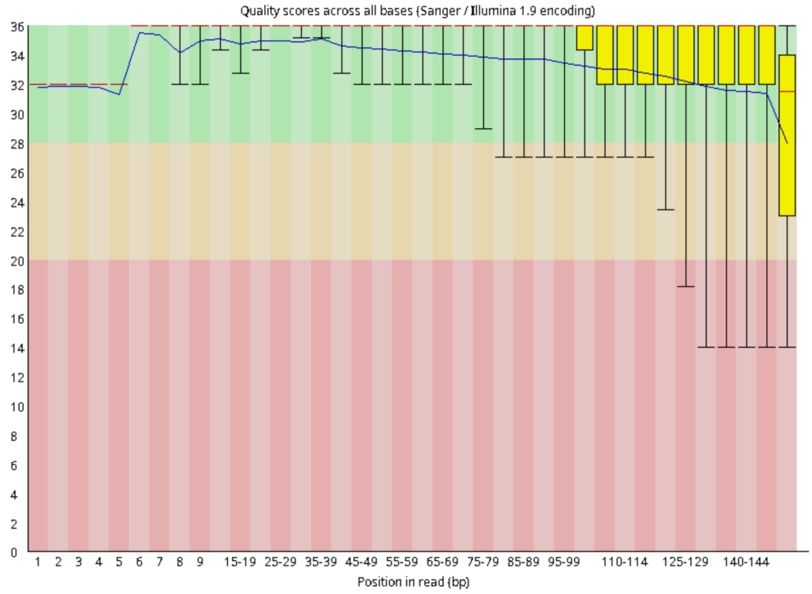
Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

Basic Statistics

Measure	Value
Filename	Erik_mat_CD4_MiLab_S52_R2_001.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1983876
Total Bases	291.2 Mbp
Sequences flagged as poor quality	0
Sequence length	35-151
%GC	41

Per base sequence quality

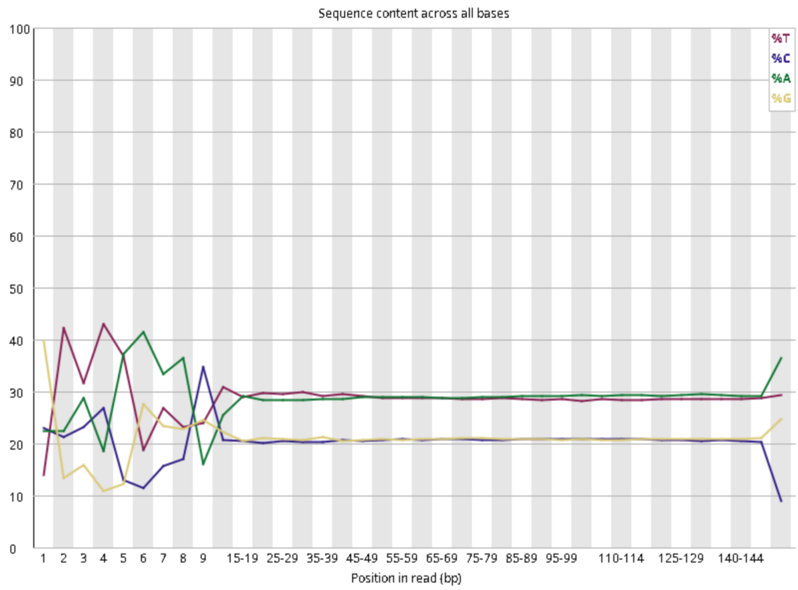


FastQC Report

Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

Per base sequence content



видимо есть какие-то адаптеры

посмотрим, что будет после тримминга

R1

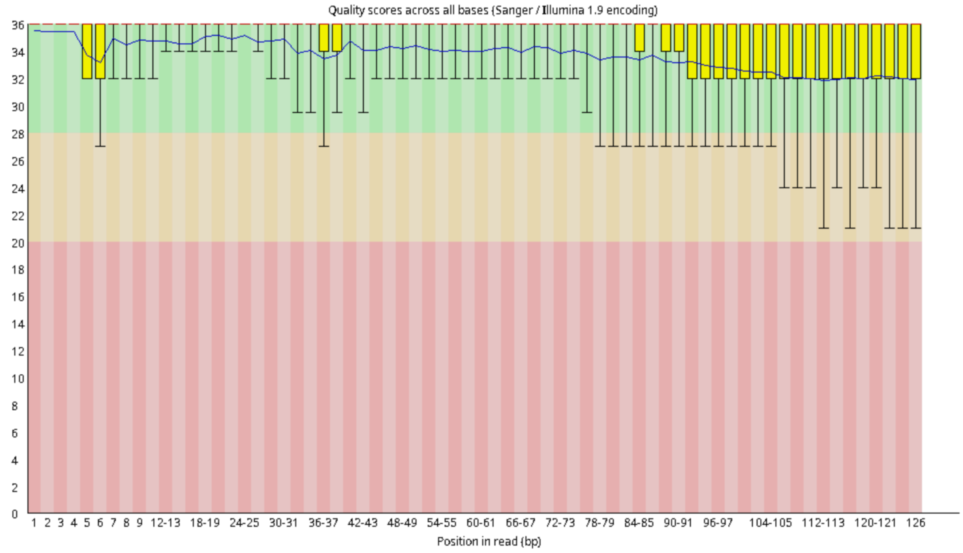
Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

Basic Statistics

Measure	Value
Filename	Erik_mat_CD4_MiLab_S52_R1_001_trimmed.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1875473
Total Bases	227.6 Mbp
Sequences flagged as poor quality	0
Sequence length	15-126
%GC	41

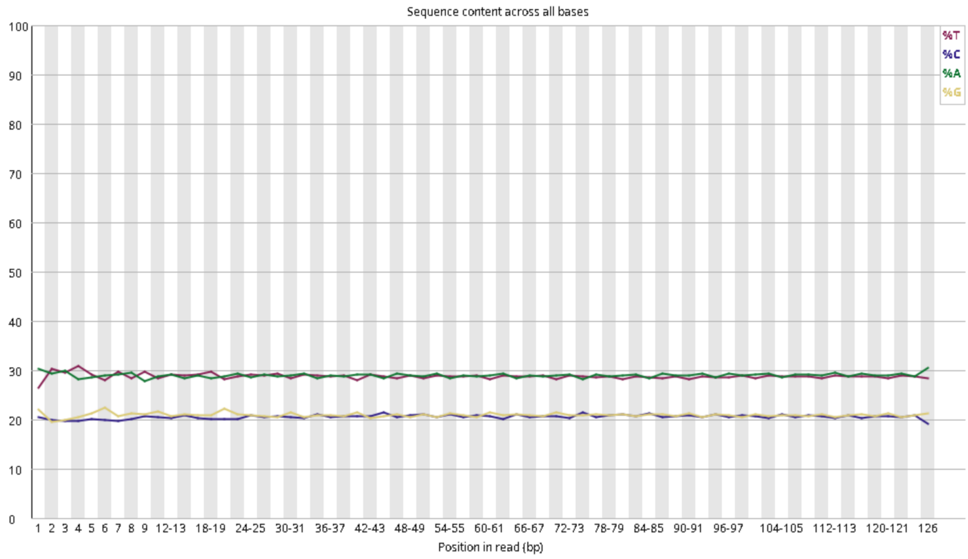
Per base sequence quality



Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

Per base sequence content



R2

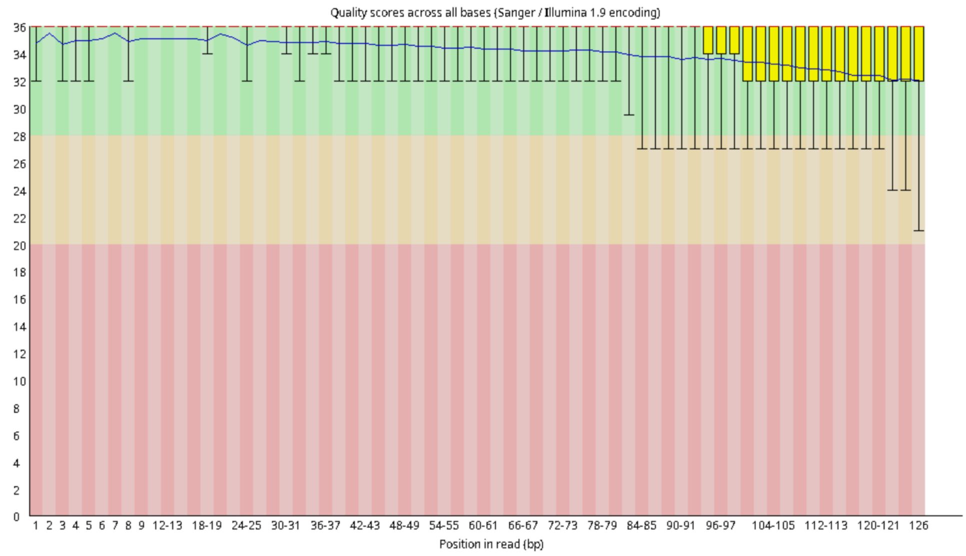
Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

Basic Statistics

Measure	Value
Filename	Erik_mat_CD4_MiLab_S52_R2_001_trimmed.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1875473
Total Bases	227.6 Mbp
Sequences flagged as poor quality	0
Sequence length	15-126
%GC	41

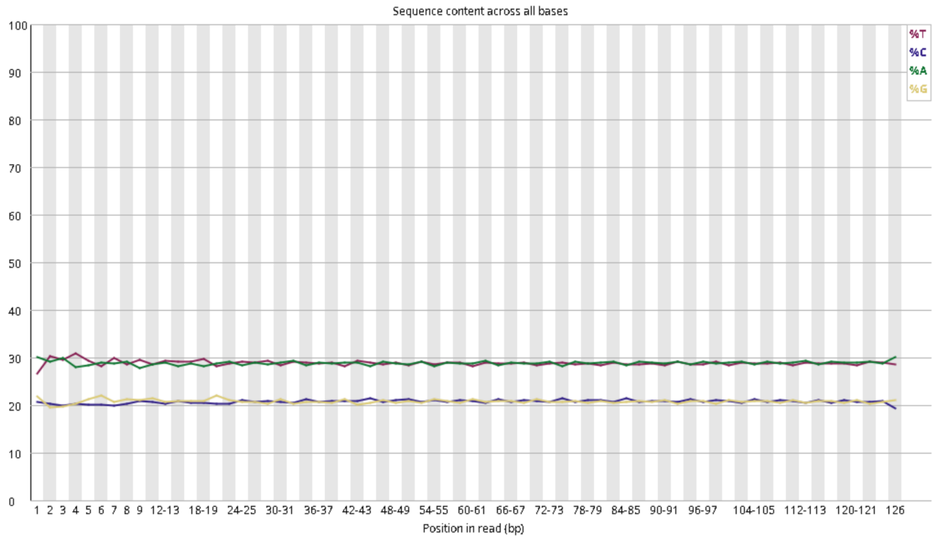
Per base sequence quality



Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

Per base sequence content



адаптеры пропали и качество улучшилось, значит берем после тримминга

что получили

```
1	StringTie	exon	633888	634233	1000	.	.	gene_id "STRG.1"; transcript_id "STRG.1.1"; exon_number "1"; cov "13.426302";
1	StringTie	transcript	7984894	7985251	1000	.	.	gene_id "STRG.2"; transcript_id "STRG.2.1"; cov "4.846369"; FPKM "15.288016"; TPM "490.393616";
1	StringTie	exon	7984894	7985251	1000	.	.	gene_id "STRG.2"; transcript_id "STRG.2.1"; exon_number "1"; cov "4.846369";
1	StringTie	transcript	8863890	8864092	1000	.	.	gene_id "STRG.3"; transcript_id "STRG.3.1"; cov "8.019705"; FPKM "25.298401"; TPM "811.496643";
1	StringTie	exon	8863890	8864092	1000	.	.	gene_id "STRG.3"; transcript_id "STRG.3.1"; exon_number "1"; cov "8.019705";
1	StringTie	transcript	8866279	8866501	1000	.	.	gene_id "STRG.4"; transcript_id "STRG.4.1"; cov "5.739910"; FPKM "18.106720"; TPM "580.809082";
1	StringTie	exon	8866279	8866501	1000	.	.	gene_id "STRG.4"; transcript_id "STRG.4.1"; exon_number "1"; cov "5.739910";
1	StringTie	transcript	26281058	26281509	1000	.	.	gene_id "STRG.5"; transcript_id "STRG.5.1"; cov "15.070796"; FPKM "47.541279"; TPM "1524.981323";
1	StringTie	exon	26281058	26281509	1000	.	.	gene_id "STRG.5"; transcript_id "STRG.5.1"; exon_number "1"; cov "15.070796";
1	StringTie	transcript	26320169	26320500	1000	.	.	gene_id "STRG.6"; transcript_id "STRG.6.1"; cov "26.692772"; FPKM "84.203156"; TPM "2700.983887";
1	StringTie	exon	26320169	26320500	1000	.	.	gene_id "STRG.6"; transcript_id "STRG.6.1"; exon_number "1"; cov "26.692772";
1	StringTie	transcript	26475112	26475949	1000	.	.	gene_id "STRG.7"; transcript_id "STRG.7.1"; cov "10.003800"; FPKM "31.557289"; TPM "1012.262939";
1	StringTie	exon	26475112	26475949	1000	.	.	gene_id "STRG.7"; transcript_id "STRG.7.1"; exon_number "1"; cov "10.003800";
1	StringTie	transcript	28508830	28509588	1000	.	.	gene_id "STRG.8"; transcript_id "STRG.8.1"; cov "8.245059"; FPKM "26.009287"; TPM "834.299744";
1	StringTie	exon	28508830	28509588	1000	.	.	gene_id "STRG.8"; transcript_id "STRG.8.1"; exon_number "1"; cov "8.245059";
1	StringTie	transcript	30732505	30733919	1000	.	.	gene_id "STRG.9"; transcript_id "STRG.9.1"; cov "17.000000"; FPKM "53.627010"; TPM "1720.193359";
1	StringTie	exon	30732505	30733919	1000	.	.	gene_id "STRG.9"; transcript_id "STRG.9.1"; exon_number "1"; cov "17.000000";
1	StringTie	transcript	32274737	32275083	1000	.	.	gene_id "STRG.10"; transcript_id "STRG.10.1"; cov "10.311239"; FPKM "32.527115"; TPM "1043.372070";
1	StringTie	exon	32274737	32275083	1000	.	.	gene_id "STRG.10"; transcript_id "STRG.10.1"; exon_number "1"; cov "10.311239";
1	StringTie	transcript	32285514	32286148	1000	.	.	gene_id "STRG.11"; transcript_id "STRG.11.1"; cov "28.196850"; FPKM "88.947815"; TPM "2853.178467";
1	StringTie	exon	32285514	32286148	1000	.	.	gene_id "STRG.11"; transcript_id "STRG.11.1"; exon_number "1"; cov "28.196850";
1	StringTie	transcript	39028723	39028940	1000	.	.	gene_id "STRG.12"; transcript_id "STRG.12.1"; cov "8.224771"; FPKM "25.945286"; TPM "832.246765";
1	StringTie	exon	39028723	39028940	1000	.	.	gene_id "STRG.12"; transcript_id "STRG.12.1"; exon_number "1"; cov "8.224771";
1	StringTie	transcript	39034392	39034600	1000	.	.	gene_id "STRG.13"; transcript_id "STRG.13.1"; cov "4.827751"; FPKM "15.229285"; TPM "488.509705";
1	StringTie	exon	39034392	39034600	1000	.	.	gene_id "STRG.13"; transcript_id "STRG.13.1"; exon_number "1"; cov "4.827751";
1	StringTie	transcript	58782767	58784049	1000	.	.	gene_id "STRG.14"; transcript_id "STRG.14.1"; cov "13.249415"; FPKM "41.795681"; TPM "1340.679688";
1	StringTie	exon	58782767	58784049	1000	.	.	gene_id "STRG.14"; transcript_id "STRG.14.1"; exon_number "1"; cov "13.249415";
```

- **Первая колонка** : `StringTie` — источник данных.
- **Вторая колонка** : Тип элемента (`transcript` или `exon`).
- **Третья–шестая колонки** : Позиции элемента на хромосоме.
- **Последняя колонка** : Дополнительные атрибуты:
  - `gene_id` : Идентификатор гена.
  - `transcript_id` : Идентификатор транскрипта.
  - `cov` : Среднее покрытие транскрипта.
  - `FPKM` : Нормированная экспрессия (Fragments Per Kilobase of transcript per Million mapped reads).
  - `TPM` : Нормированная экспрессия (Transcripts Per Million).

Каждый транскрипт имеет связанные экзоны с одинаковым transcript_id. Метрики экспрессии показывают уровень выражения транскриптов в выборке.